



Customer Shopping Behaviour Analysis

Transactional data insights across 20 purchases — spending patterns, segments & product preferences

Dataset at a Glance

20

Rows

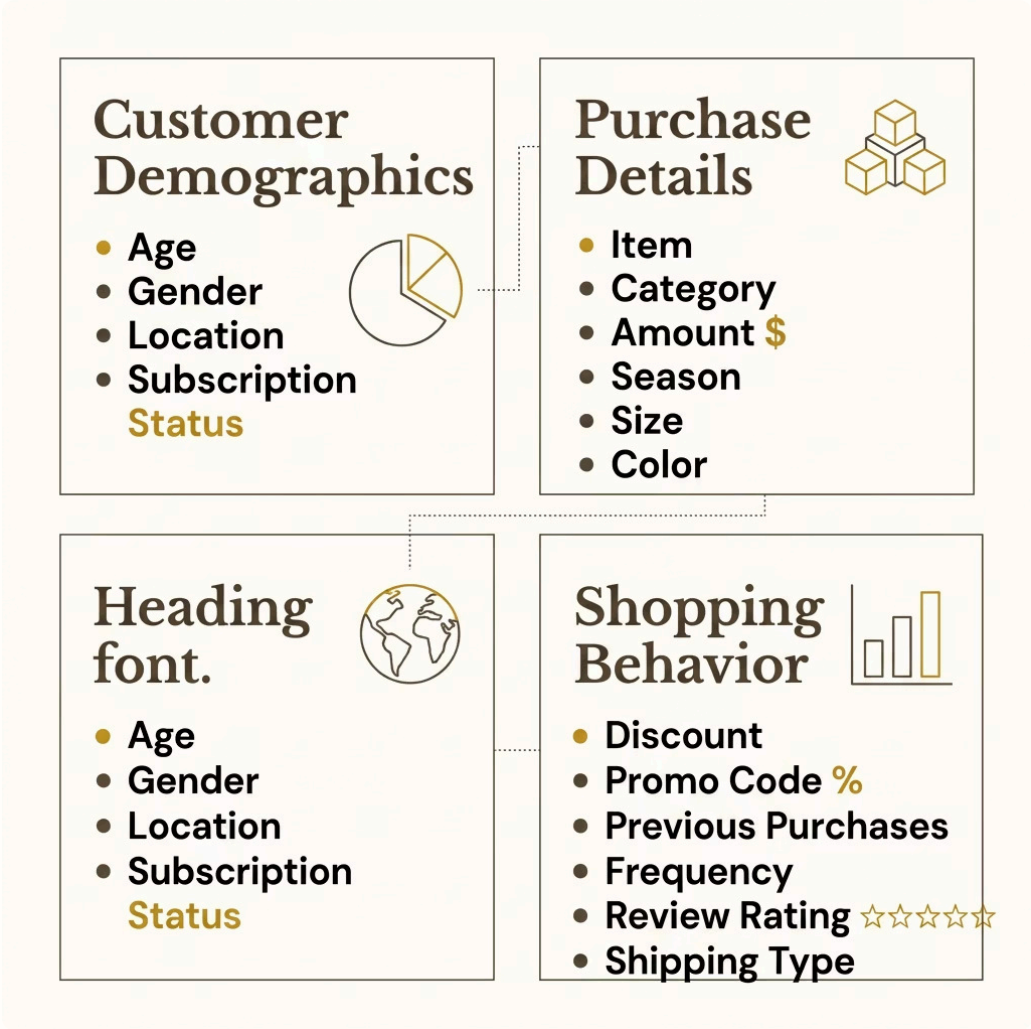
18

Columns

1

Missing Value

Review Rating



Key Feature Groups

- Demographics & subscription status
- Purchase details & categories
- Shopping behavior & frequency

Python: Data Preparation

01

Load & Explore

pandas: `df.info()`, `df.describe()`

02

Clean Missing Data

Imputed Review Rating via category median

03

Standardize & Engineer

Snake case columns; created `age_group` & `purchase_frequency_days`

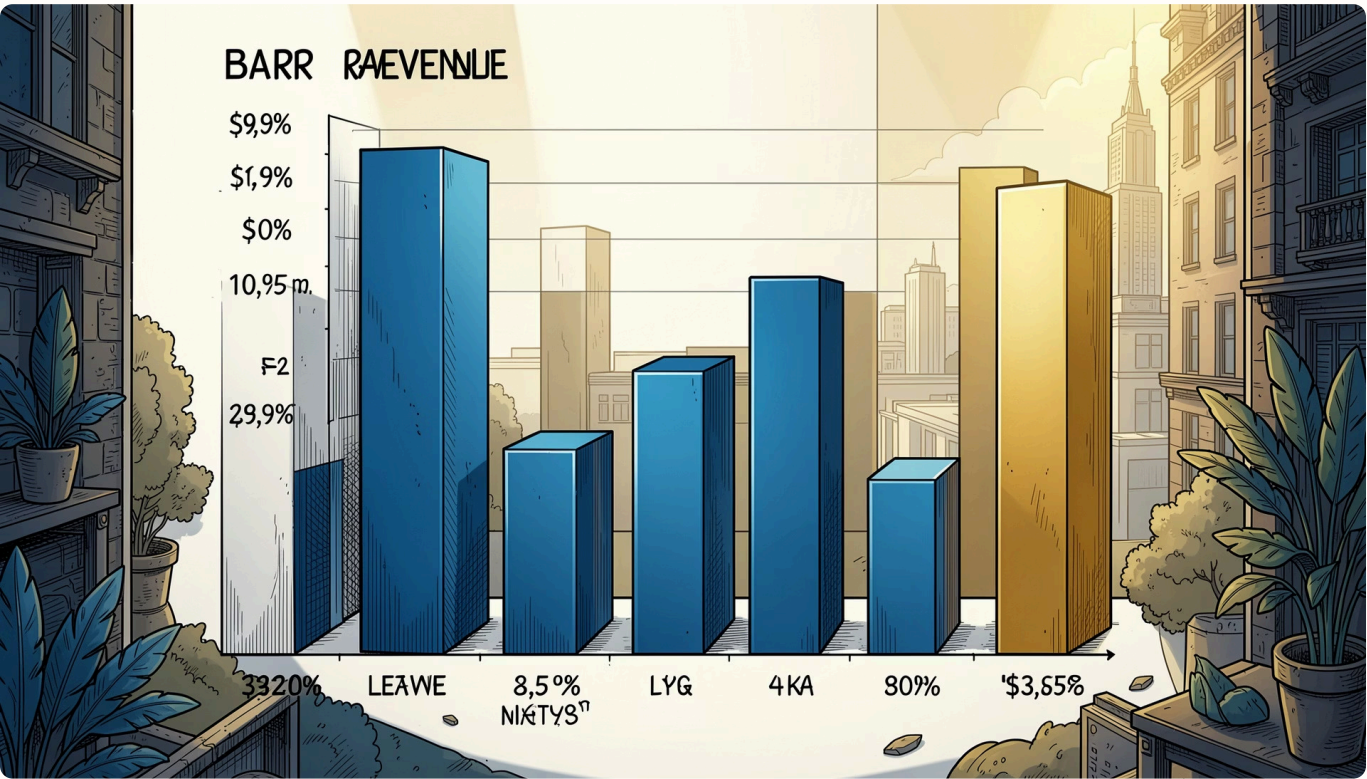
04

Load to PostgreSQL

Cleaned DataFrame loaded for SQL analysis



SQL Analysis: Revenue & Promo Insights



Revenue by Gender

Compared total revenue generated by male vs. female customers.

High-Spending Promo Users

Identified customers who used promo codes but still spent **above average** purchase amount.

SQL Analysis: Ratings & Shipping

Top 5 Products by Rating

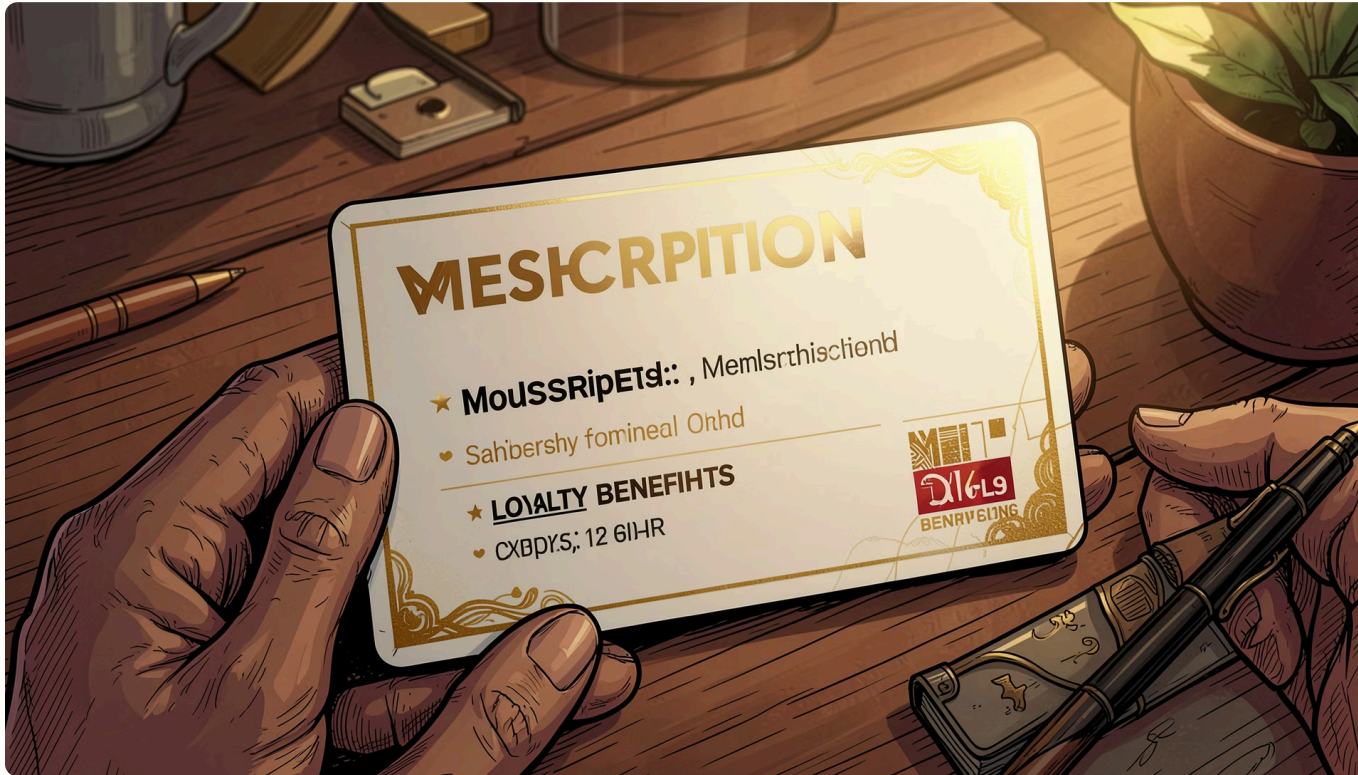
Highest average review ratings across all categories.

Shipping Type Comparison

Average purchase amount: **Next Day** vs. **Express** shipping.



SQL Analysis: Subscribers & Discounts



Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status.

Discount Applied Products

Products with the **highest percentage** of discount applied.

SQL Analysis: Customer Segmentation



Segments Defined

New

Returning

Loyal

Classified by purchase history.

SQL Analysis: Products & Age Groups

Top 3 Products per Category

Most purchased products within each category.

Repeat Buyers & Subscriptions

Customers with **>5 purchases** — are they more likely to subscribe?

Revenue by Age Group

Total revenue contribution of each age group.



Power BI Dashboard

Interactive visual dashboard presenting all key insights — revenue, segments, ratings, and trends in one view.



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → "Loyal" segment



Review Discount Policy

Balance sales boosts with margin control



Product Positioning

Highlight top-rated & best-selling products



Targeted Marketing

Focus on high-revenue age groups & express shipping users